

Sulfate-dominant apatite in the deep arc crust indicates that high oxidation state promotes copper fluxing in arcs

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ABSTRACT

Sulfide saturation during arc magma evolution depletes ascending magmas in chalcophile elements (e.g., Cu) and promotes metal recycling into the mantle. Although sulfide saturation is posited to be delayed by high oxygen fugacity (fO_2) in the deep crust of arcs, the timing and controls on sulfide saturation in arcs remain disputed. Here, we report new constraints on S speciation in the deep arc crust using measurements of the relative abundance of S species (S^{2-} , S^{4+} , and S^{6+}) in apatite in deep crustal arc cumulates from the Sierra Valle Fértil–La Huerta (Argentina). Sulfide is rare in primitive cumulates ($<0.01\%$), only becoming abundant in evolved cumulates ($Mg\# < 0.65$), indicating that sulfide saturation was delayed in the magmatic evolution. Apatite yields consistently high $S^{6+}/\Sigma S$ (0.8–1.0), giving an fO_2 of 1.4 ± 0.4 log units above the fayalite-magnetite-quartz (FMQ) redox buffer. Delayed sulfide saturation was therefore the result of the parental magma having elevated fO_2 acquired from its mantle source. We use modeling to show that magma differentiation at FMQ+1.4 would be sufficient to delay sulfide saturation and produce mafic to intermediate melts with elevated Cu concentrations. Oxidized, hydrous intermediate magmas may therefore play a key role in delivering elevated sulfur and Cu fluxes to magmatic-hydrothermal systems and the atmosphere, contributing to the Cu deficit of modern continental crust.

INTRODUCTION

The speciation of sulfur in arc magmas controls the transport of Cu through the crust to be vented to the atmosphere or to form magmatic-hydrothermal ore deposits. At the oxygen fugacity (fO_2) typical of arc magmas (+0.6 to +1.9 log units relative to the fayalite-magnetite-quartz redox buffer [FMQ]; Cottrell et al., 2021), most melts contain both sulfide (S^{2-}) and sulfate (S^{6+}) species, unlike mid-ocean-ridge basalt (MORB), which contains exclusively S^{2-} (Jugo et al., 2010). Trace-element systematics of arc lavas and cumulates indicate that most arc magmas become saturated in sulfide during their differentiation, which may produce a reservoir of chalcophile metals (e.g., Cu, Au) in the deep crust (Jenner, 2017; Chen et al., 2020). The timing of this sulfide saturation is closely related

to S speciation (and therefore fO_2) and initial S concentrations in magmas (O'Neill and Mavrogenes, 2022). Early sulfide saturation (i.e., at high melt $Mg\#$) in less oxidized or S-rich magmas would promote the formation of Cu-rich cumulates that can eventually be recycled into Earth's mantle (Chen et al., 2020). By contrast, late sulfide saturation (i.e., at lower melt $Mg\#$) in oxidized or S-poor magmas would favor the ascent of Cu-rich magmas and the fluxing of Cu to Earth's atmosphere and to magmatic-hydrothermal systems (Cox et al., 2020).

Although arc magmas are more oxidized than MORB (Cottrell et al., 2022), whether this is inherited from their sources or is engendered by crustal differentiation remains disputed and has key implications for sulfur behavior (Parkinson and Arculus, 1999; Tang et al., 2020; Holycross and Cottrell, 2023). It has been suggested that primary arc magmas have a similar fO_2 to MORB and may acquire their higher fO_2 values during crustal differentiation (Lee et al.,

2012). By contrast, studies of Fe and S speciation in primitive arc magmas indicate that arc magmas acquire their elevated fO_2 from their mantle sources and are sulfate-dominated before entering the crust (Parkinson and Arculus, 1999; Muth and Wallace, 2021). However, some experiments indicate that sulfide stability is enhanced with increasing pressure, and even the most oxidized arc magmas in the deep crust will be sulfide saturated (Matjuschkin et al., 2016). Other experimental studies argue against the strong pressure effect on S^{6+}/S^{2-} , suggesting a stronger control of temperature and/or composition (Nash et al., 2019; O'Neill and Mavrogenes, 2022). Clearly, the interplay among fO_2 , sulfur speciation, and sulfide saturation during the deep crustal evolution of arc magmas remains disputed and requires further study.

Here, we report $S^{6+}/\Sigma S$ values measured in apatite from a suite of cumulates with variable sulfide abundance from the deep to midcrustal section of the Sierra Valle Fértil–La Huerta (VFLH), Argentina. We show that during differentiation, these magmas had consistently high $S^{6+}/\Sigma S$, and thus acquired high fO_2 directly from the source. We find that this high fO_2 delayed sulfide saturation in the deep crust, which likely facilitated the ascent of mafic-intermediate melts that escaped early Cu depletion in the deep crust.

STUDY AREA

The VFLH is the southernmost expression of magmatism related to the Ordovician Famatinian arc (Otamendi et al., 2009). The system shows a general SW to NE transition from mafic to intermediate to silicic rocks (Fig. 1) and was emplaced at lower to midcrustal depths equivalent to pressures of 8–5 kbar (Tibaldi et al., 2013). The deepest, mafic unit consists of predominantly gabbro-norites and diorites with

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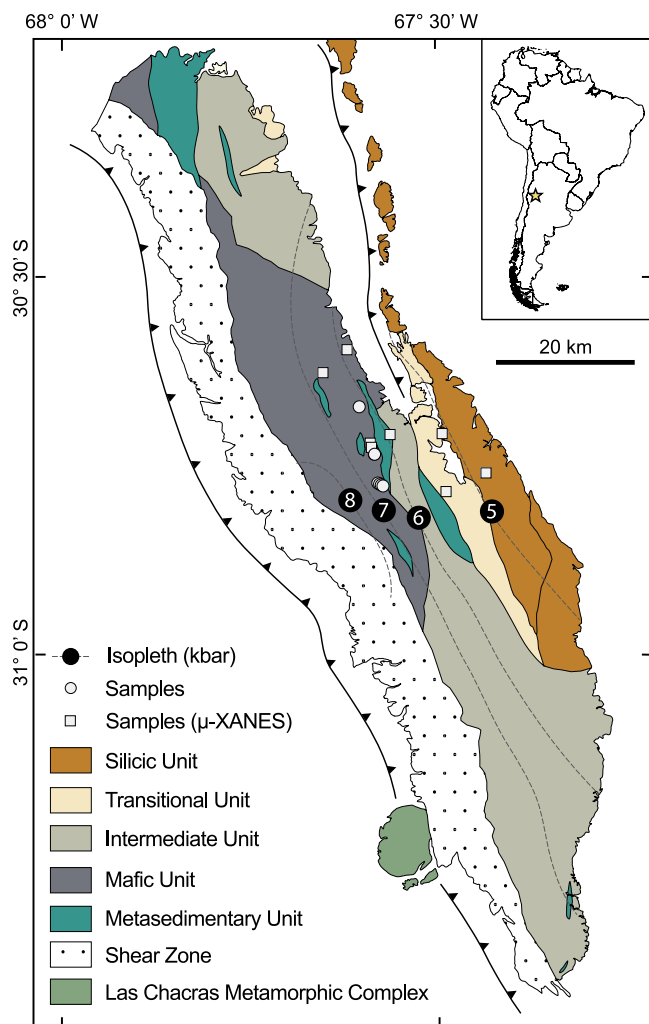


Figure 1. Simplified geologic map of Sierra Valle Fértil–La Huerta (VFLH; after Giovanini et al., 2024) showing key geologic units in crustal section, samples studied, and estimated pressures and isobars (Camilletti et al., 2020). Inset: Location of study area in South America. μ -XANES—micro-X-ray absorption near edge spectroscopy.

subordinate ultramafic-mafic bodies. The intermediate unit consists of heterogeneous amphibole- and biotite-bearing diorites and tonalites. The uppermost silicic unit is comprised of large-scale granodiorite and monzogranite intrusions, underlain by a transitional unit (Otamendi et al., 2009). The textures of samples indicate they

are cumulate dominated, and their bulk compositions follow the compositional evolution of cumulates in fractional crystallization experiments at deep crustal pressures (Fig. 2A; Nandedkar et al., 2014). Isotopic data indicate these magmas were predominantly mantle-derived materials, with evidence for source contami-

nation and limited assimilation (Cornet et al., 2022). This magmatic sequence is ideally suited to study the evolution of sulfur in the deep to middle arc crust due to the ranges of depths and differentiation levels that are exposed.

LATE APPEARANCE OF SULFIDES IN CUMULATES

Sulfides are present in almost all investigated samples, yet they occur in variable modal abundances and compositions. Sulfide is rare (<0.01 vol%) and typically interstitial in the most primitive cumulates and is dominated by chalcopyrite with lesser pyrrhotite and rare pentlandite, indicating near-solidus saturation of minor Cu-rich sulfide liquid (SL; Yund and Kullerud, 1966). In more evolved gabbronorites, sulfide can be present up to 0.24 vol% abundance, as an interstitial phase and included in phenocryst phases. These phases are predominantly pyrrhotite with lesser chalcopyrite, indicating they were trapped as monosulfide solid solution (MSS; Yund and Kullerud, 1966). In intermediate rocks, sulfide can be less abundant and less stable with increasing SiO_2 , with common Fe-oxide rims and an increasing abundance of pyrite relative to other sulfides. Overall, sulfide only becomes a significant phase (>0.08 vol%) in more evolved cumulates (<0.65 Mg#), indicating that the primitive magmas were sulfide-undersaturated until they cooled and differentiated (Fig. 2B). Comparison of these cumulate compositions with experimental cumulate-melt pairs (Fig. 2A; Nandedkar et al., 2014) indicates that sulfide-undersaturated cumulates at VFLH were in equilibrium with mafic-intermediate melts that could have ascended the crust without reaching Cu-depleting sulfide saturation (Fig. 2A).

DOMINANCE OF SULFATE IN CUMULATE APATITE

The late sulfide saturation in the VFLH (at cumulate Mg# ~0.65) contrasts with early and

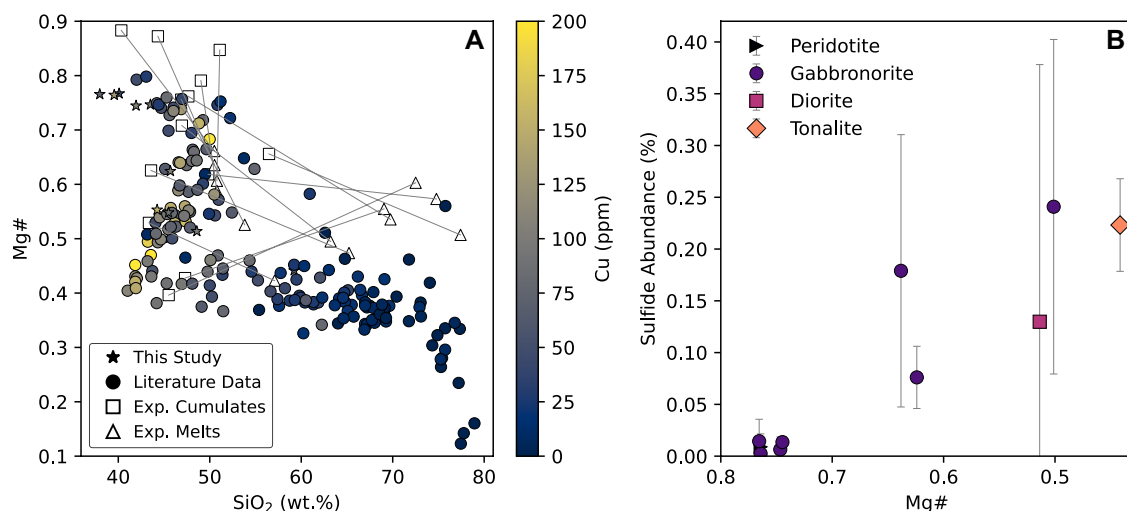


Figure 2. (A) Bulk-rock Mg# vs. SiO_2 for Sierra Valle Fértil–La Huerta (VFLH) rocks from literature sources and this study colored according to bulk-rock Cu concentration. Upper scale bound is restricted to 200 ppm to suppress effect of outliers. Experimental (Exp.) cumulate compositions are from fractional crystallization experiments for arc magmas at 0.7 GPa (Nandedkar et al., 2014). (B) Abundance of sulfide in samples at different bulk-rock Mg# values (see Supplemental Material [see text footnote 1]).

abundant sulfide saturation in many intraplate settings in cumulates with $Mg\# > 0.8$ (Holwell et al., 2022; Kim et al., 2023) and in some arc systems, such as the Talkeetna crustal section (Rezeau et al., 2023). Late sulfide saturation has been documented in other arc cumulates (Chang and Audétat, 2018; Chen et al., 2020) and inferred from chalcophile systematics at some arc volcanoes (Cox et al., 2020). This delay in sulfide saturation may be due to the high fO_2 at which these magmas differentiated, reducing S^{2-} activity in the melt at a given S concentration. Therefore, all else being equal, more differentiation would be required to elevate melt S^{2-} to reach sulfide saturation. Alternatively, initially low bulk S contents in the primitive magma would also require more crystallization (i.e., passive enrichment of S in the melt) to trigger sulfide saturation.

To test whether initially high fO_2 delayed sulfide saturation in these cumulates, we analyzed sulfur speciation in apatite from eight samples by micro-X-ray absorption near edge spectroscopy (μ -XANES) at the S K-edge (see the Supplemental Material¹). Apatite can incorporate three valence states of sulfur into its structure (S^{2-} , S^{4+} , and S^{6+}) with the ratio of S^{6+} relative to other sulfur species ($S^{6+}/\Sigma S$) in apatite increasing steeply with fO_2 over the range from approximately FMQ to $\sim$ FMQ+1.2 (Konecke et al., 2019). If primitive arc magmas have an initial fO_2 close to FMQ and become progressively oxidized with crustal differentiation, an increase in apatite $S^{6+}/\Sigma S$ should be observed through the VFLH rocks. Alternatively, if apatite $S^{6+}/\Sigma S$ is consistently high, this would indicate the magmas acquired their high fO_2 in the mantle source.

We report 49 successful averaged μ -XANES spectra of apatite from eight samples of VFLH cumulates ranging from gabbronorite to granodiorites (Fig. 3A). All spectra show strong S^{6+} peaks, weak S^{4+} peaks, and rare S^{2-} peaks. Calculated peak area ratios, reported as $S^{6+}/\Sigma S$, are remarkably homogeneous ($S^{6+}/\Sigma S = 0.92 \pm 0.04$) and do not vary strongly with crustal depth or differentiation (Fig. 3A). We used these values to quantify magmatic fO_2 using the S-in-apatite oxybarometer (Konecke et al., 2019), with a Monte Carlo uncertainty propagation (see Supplemental Material). This method returned a mean fO_2 of FMQ+1.4 $\pm$ 0.4 (Fig. 3B), falling within the typical range of arc magmas (Cottrell et al., 2022).

OXIDIZED PRIMITIVE ARC MAGMAS DELAY SULFIDE SATURATION

The presence of sulfide in these rocks combined with the abundance of sulfate in apatite

indicate that the magmatic system was within the fO_2 range of the sulfide to sulfate transition (FMQ+0.5 to FMQ+2.0; Jugo et al., 2010). Because S^{2-} is significantly less compatible in apatite than S^{6+} (Konecke et al., 2019), the melt $S^{6+}/\Sigma S$ may have been sufficiently high for negligible S^{2-} incorporation in apatite, yet it was sufficient to delay sulfide saturation to late in the cumulate line of descent. Our data indicate that these magmas were oxidized from relatively early in their magmatic history, likely acquiring their high fO_2 and high $S^{6+}/\Sigma S$ values during mass transfer of oxidized components from the subducting slab (Kelley and Cottrell, 2009; Muth and Wallace, 2021) as opposed to during crustal differentiation (Tang et al., 2020), consistent with isotopic evidence for source contamination in VFLH magmas (Cornet et al., 2022).

Delayed sulfide saturation during deep crustal magma differentiation could limit loss of Cu to sulfide-bearing cumulates, producing melts of mafic-intermediate composition that may deliver elevated Cu fluxes to the atmosphere and magmatic-hydrothermal systems. To test the potential for high magmatic fO_2 to delay sulfide saturation and produce elevated Cu in evolved magmas, we developed a redox-sensitive sulfide saturation model. Our model computes residual melt compositions through an isobaric Gibbs free energy minimization using the thermodynamic model of Holland et al. (2018) in MAGEMin (Riel et al., 2022). Our model simulates fractional crystallization of a primitive arc magma composition from 1200 °C to 700 °C at a randomly drawn pressure appropriate for the deep to midcrust (8–5 kbar). In each simulation, fO_2 is buffered at a randomly drawn value to cover the range of arcs and MORB (FMQ to FMQ+2.5) to explore the fO_2 control on sulfide saturation. Sulfur speciation (S^{6+}/S^{2+}) in the residual melt is then computed using an experimentally calibrated model (O'Neill and Mavrogenes, 2022), and the sulfur content required for sulfide saturation (SCSS) is calculated (Li and Zhang, 2022) using pySulfSat (Wieser and Gleeson, 2023), with the initial bulk S and Cu contents randomly selected from the range observed in primitive arc magmas (1000–3000 and 50–100 ppm, respectively).

In total, 1000 simulations indicated that sulfide saturation occurs at near-liquidus conditions for those computed between FMQ and FMQ+1 (Fig. 4A), in alignment with observations from intraplate, less oxidized settings (Holwell et al., 2022; Kim et al., 2023). Sulfide saturation is progressively driven from the liquidus to near solidus as fO_2 increases from FMQ+1 to FMQ+2.5, due to more crystallization being required to elevate melt S^{2-} to concentrations required for sulfide saturation. These results indicate that the fO_2 predicted from our apatite would be sufficient to delay sulfide saturation beyond the liquidus (Fig. 4A). Our model also

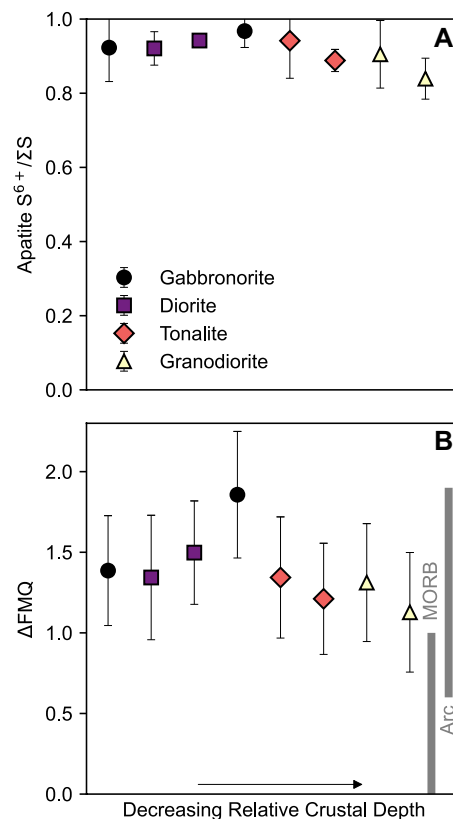


Figure 3. (A) Averaged apatite $S^{6+}/\Sigma S$ peak ratio determined by micro-X-ray absorption near edge spectroscopy (μ -XANES) for each sample. Samples are ranked on x-axis according to their position relative to mapped isobars (Camilletti et al., 2020). Errors indicate 2σ on $S^{6+}/\Sigma S$ per sample. (B) Log units relative to fayalite-magnetite-quartz redox buffer (ΔFMQ) calculated for each sample using apatite S oxybarometer (Konecke et al., 2019) showing mean and standard deviation from Monte Carlo uncertainty propagation. Vertical bars show range of oxygen fugacity (fO_2) in mid-ocean-ridge basalt (MORB) and arc magmas (Cottrell et al., 2022).

indicates that lower bulk S contents in the primitive magma can delay sulfide saturation yet have a subordinate effect relative to fO_2 .

IMPLICATIONS FOR COPPER CYCLING IN ARCS

We computed the Cu concentrations of differentiating melts in our simulations using SL and MSS/melt partitioning depending on the temperature at which sulfide forms. The results show that Cu behaves compatibly in simulations buffered at fO_2 between FMQ and FMQ+1 (due to sulfide precipitation) but behaves incompatibly in more oxidized simulations until the point of sulfide saturation, which then causes a Cu decrease in the residual melt (Fig. 4B). This suggests that mafic-intermediate melts in equilibrium with Valle Fertile ultramafic cumulates may have had elevated Cu concentrations (100–200 ppm), in accordance with the incom-

¹Supplemental Material. Analytical and modeling methods, Figures S1–S12, and Tables S1–S2. Please visit <https://doi.org/10.1130/GEOLOGY.S30505751> to access the supplemental material; contact editing@geosociety.org with any questions.

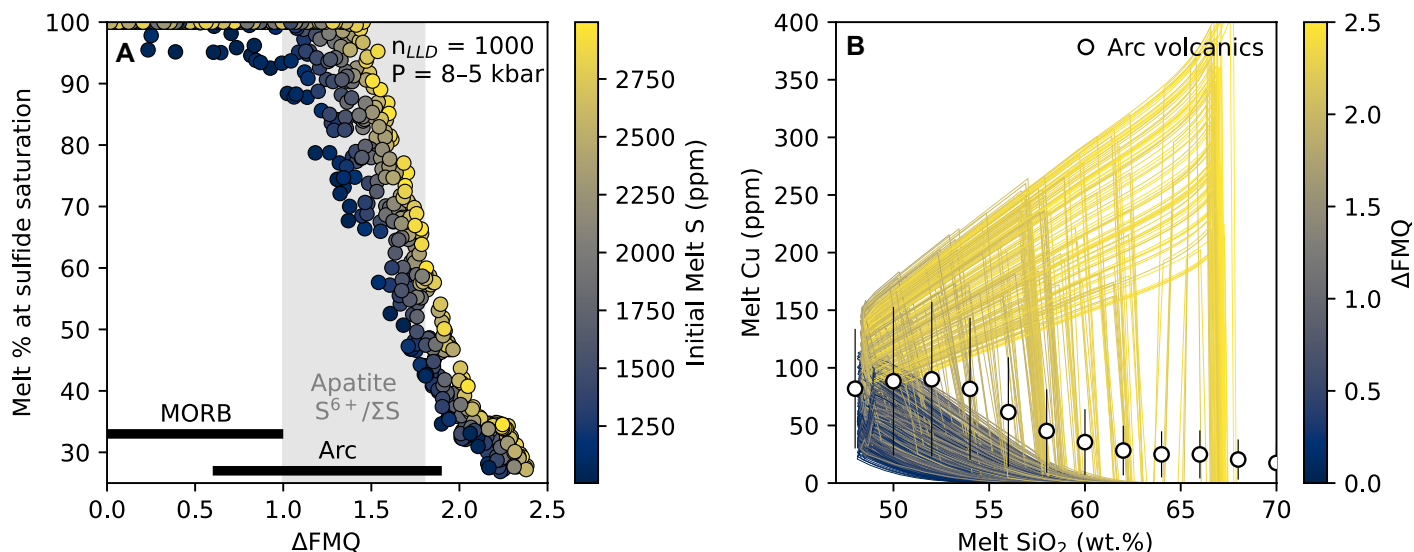


Figure 4. Results of 1000 sulfide saturation simulations. (A) Melt % at which sulfide saturates in each simulation as function of log units relative to fayalite-magnetite-quartz redox buffer (ΔFMQ), colored by starting bulk S content of simulation. Horizontal black bars indicate approximate range of oxygen fugacity ($f\text{O}_2$) for mid-ocean-ridge basalt (MORB) and arc magmas (Cottrell et al., 2022); gray shaded field indicates Sierra Valle Fértil–La Huerta (VFLH) $f\text{O}_2$ predicted from apatite S oxybarometer. **(B)** Liquid line of descent of melt Cu colored by ΔFMQ , which is used to buffer $f\text{O}_2$. Symbols show bulk-rock data for arc volcanic rocks from Geochemistry of Rocks of the Oceans and Continents (GEOROC) database (<https://georoc.eu/georoc/new-start.asp>), and error bars indicated their 95th percentile.

patible behavior of Cu with differentiation in some mafic-intermediate arc volcanoes (Cox et al., 2020; Grondahl and Zajacz, 2022). However, our model indicates that delaying sulfide saturation to the intermediate-felsic compositions typical of porphyry Cu deposits is challenging and only possible in simulations with $f\text{O}_2$ beyond that typically encountered in arcs ($\sim\text{FMQ}+2.5$; Fig. 4B). The documentation of sulfide-bearing cumulate xenoliths in porphyry Cu deposits has been used to argue against sulfide saturation being detrimental to porphyry ore formation (Du and Audétat, 2020). Yet, oxidized, volatile-rich magmas may attain fluid saturation (and/or anhydrite saturation where $\Delta\text{FMQ} > 1.8$; Kleinsasser et al., 2022) earlier than sulfide saturation, which can additionally inhibit loss of Cu to deep cumulates and promote the delivery of enhanced sulfur and Cu fluxes to magmatic-hydrothermal systems.

The incompatible nature of Cu during oxidized differentiation is not typical in arc volcanic rocks (Fig. 4B) and is only observed at a small number of arc volcanoes (Cox et al., 2020). Because the most typical products of arc volcanoes are oxidized ($>\text{FMQ}+1$) and hydrous basalts and andesites, fluid exsolution prior to sulfide saturation may be common underneath arc volcanoes, promoting transfer of sulfur and Cu to a fluid phase, which is vented to the atmosphere. This is consistent with the significantly higher sulfur and metal (including Cu) fluxes of arc volcanoes relative to less oxidized, dry magmas in other settings (Wallace and Edmonds, 2011; Edmonds et al., 2018) and may lead to the bulk-rock record of high Cu trends in volcanic rocks being underrepresented. In the case

of VFLH, the paucity of sulfide and lack of significant Cu enrichment ($\text{Cu} < 100$ ppm) in the deepest cumulates suggest that delamination and recycling of the deepest crust into the mantle may not be dominant processes in producing the depletion of Cu in the continental crust relative to primitive arc magmas (Jenner, 2017). Instead, the ascent and degassing of oxidized and hydrous arc magmas may be important mechanisms in the transport of sulfur and metals among Earth's mantle, crust, and atmosphere.

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